H1: a homogeneous upstream candidate

13 October 2026 — analytical qualification, not an adopted traffic contract

1. Traffic meaning and assumption boundary

The ending lane E seeks early transfer into adjacent continuing lane A; continuing lane B does not receive lateral traffic. A finite uniform footprint initially occupies $[0, \ell]$ inside a fixed preparation area $[0, L]$, with $0 < \ell \leq L$. At the decision instant it has densities e_0, a_0, b_0 and mass $M = \ell(e_0 + a_0 + b_0) > 0$. Initial symmetry means $a_0 = b_0$, not $a(t) = b(t)$ after merging. No new arrivals are included. All quantities and the initial footprint are fixed across action comparisons.

One constant common advice $v \in [v_-, v_+]$, $v_- > 0$, is realized as the preparation speed on the admitted free-flow branch. This is a declared analytical response hypothesis, not established compliance. Lane changes are represented by a uniform same-coordinate lateral source. The model does not reproduce the diagonal next-cell movement of the native CTM algorithm.

The physical interaction scale $h > 0$ is fixed across actions. It converts a throughput-equivalent opportunity into a lateral density-change rate. Its meaning is repeated physical opportunity per unit progress, rather than an arbitrary numerical tick. It is an explicit adaptation requiring provenance or robustness qualification; no source-derived or fitted value is adopted. The early work weight $\gamma \geq 1$ and late work weight $\alpha \geq 1$ are distinct. Work is normalized so that an ordinary mass unit uses one work unit.

Quantity	Meaning	Units
L, ℓ, h, z	geometry, footprint, opportunity scale, progress	m
v, w	common free speed, backward-wave magnitude	m/s
k_j, e, a, b	jam or lane density	veh-eq/m
q_E, q_A, q_B	ordinary longitudinal mass flux	veh-eq/s
C, S	receiving-work rate (ordinary-work normalization)	work/s
f	lateral mass-flux equivalent at a representative opportunity	veh-eq/s
$m = f/h$	local lateral density source	veh-eq/(m s)
γ, α	receiving work per lateral/late mass unit	dimensionless

The boundary must accept approaching traffic for the no-wait transport qualification below. A waiting terminal is considered as a reopen condition, not silently added to the equations.

2. State to receiving, own demand, and completed merging

Use one declared triangular diagram for the diagnostic family:

$$Q_v(k) = \min\{vk, w(k_j - k)\}, \quad k_c(v) = \frac{wk_j}{v + w}, \quad C(v) = vk_c(v), \qquad w, k_j > 0.$$

For density a the adjacent receiving rate is

$$S(v, a) = \min\{C(v), w(k_j - a)\}.$$

Restrict the initial densities over the whole action range by

$$0 \leq e_0, a_0, b_0 \leq k_c(v_+). \tag{A}$$

The candidate will keep $a \le k_c(v)$ and $e \le e_0$, so ordinary traffic remains on the linear free-flow branch. Own longitudinal flow and the declared rate analogue of eager donor availability are

$$q_A = va, \quad q_E = ve, \quad q_B = vb_0, \qquad D_E = ve.$$

Within this branch $S = C$ and $q_A \le S$. The local R1-inspired rate allocation is

$$f(v, e, a) = \min\left\{D_E, \frac{S(v, a) - q_A}{\gamma}\right\}, \qquad m = \frac{f}{h}. \tag{1}$$

Consequently

$$0 \le f \le ve, \qquad va + \gamma f \le S(v, a), \qquad \dot{e} = -m, \quad \dot{a} = m, \quad \dot{b} = 0. \tag{2}$$

Actual transferred mass is $m\,dx\,dt$; it is not $\gamma m\,dx\,dt$. No clipping rule is needed inside the admitted domain because $S - q_A \ge 0$.

Adaptation boundary. Carey's R1 supports own-lane-first residual receiving and the gap-weighted work inequality. Its native request is a cell stock in a native interval. Equations (1)–(2) additionally choose a progress-based rate offer, same-coordinate transfer and a physical scale h. They are not a verbatim continuous-time transcription of Algorithm A1. Here q_E describes road-frame longitudinal motion; f is the opportunity-equivalent bound used in a separate lateral source. It is not a second longitudinal outflow to add to q_E at the same boundary. A native diagonal donor sending budget cannot be imported unchanged into this altered representation.

Likewise, C at successive road cross sections is not one pooled capacity to sum and spend remotely. Uniformity makes the local budget identical at every occupied location; the choice f/h specifies lateral intensity locally. This is precisely the additional homogeneous interaction assumption to qualify.

3. Why the homogeneous member is invariant

For an analytical transport reference, use the proposed balance equations within the occupied region:

$$\partial_t e + v\partial_x e = -m(e, a; v), \qquad \partial_t a + v\partial_x a = m(e, a; v), \qquad \partial_t b + v\partial_x b = 0. \tag{3}$$

Outside the common footprint all three densities are zero and the source is zero. Equal initial values at all occupied positions produce equal solutions of the same ordinary differential equations along their characteristics. Thus the occupied body stays uniform while its edges translate at v. Equation (3) is a reference proof, not a position-resolved simulator or an assertion about congested waves in this task.

Put $r = e_0 + a_0 = e + a$. In the free-flow member (2) becomes

$$\dot{e} = -\frac{v}{h}\min\left\{e, \frac{k_c(v) - r + e}{\gamma}\right\}, \qquad a = r - e. \tag{4}$$

At $e = 0$ the rate is zero. The donor-bound $m \le ve/h$ preserves nonnegative ending density. At $a = k_c(v)$ the spare receiving vanishes, so the adjacent density cannot cross that boundary. Therefore

$$0 \le e(t) \le e_0, \qquad a_0 \le a(t) \le \min\{r, k_c(v)\}, \qquad b(t) = b_0. \tag{5}$$

Condition (A) and this invariant ensure the selected common speed is feasible throughout the candidate upstream motion. They do not prove that every traffic state under a VSL actually lies in this domain.

4. The opportunity clock and exact branch references

With physical progress $z = vt$ and $\delta(v) = k_c(v) - r$, (4) gives

$$\frac{de}{dz} = -\frac{1}{h} \min\left\{e, \frac{\delta(v) + e}{\gamma}\right\}. \tag{6}$$

The explicit factor v cancels. If normalized receiving $C(v)/v$ is unchanged between actions, this candidate has the same preparation per unit distance. Additional elapsed time alone supplies no benefit.

For a donor-limited branch starting at (z_s, e_s),

$$e(z) = e_s \exp\left[-\frac{z - z_s}{h}\right]. \tag{7}$$

For a residual-service branch,

$$e(z) = (e_s + \delta) \exp\left[-\frac{z - z_s}{\gamma h}\right] - \delta. \tag{8}$$

Use these formulas only while the corresponding minimum branch remains active. For $\gamma > 1$ and $\delta > 0$, the switch threshold is $e = \delta/(\gamma - 1)$. If $\delta < 0$, the donor cannot vanish while the target stays free flow; its limiting residual is $-\delta = r - k_c(v)$. These are exact analytical branch references, not numerical rollouts.

In the triangular diagnostic

$$k_c'(v) = -\frac{wk_j}{(v + w)^2} < 0.$$

For a service branch with fixed initial branch data,

$$\frac{\partial e}{\partial \delta} = \exp\left[-\frac{z - z_s}{\gamma h}\right] - 1 \leq 0.$$

Slower speed can therefore increase preparation via changed normalized density headroom. This conditional preparation sensitivity is not the derivative of complete time with a waiting terminal. Switch events and any new boundary need their own derivative qualification later.

5. Compact physical arrivals and whole-cohort conservation

The front reaches L at $t_f = (L - \ell)/v$, and the rear at $t_r = L/v$. For $0 \leq t < t_r$, the length still in preparation is

$$s(t) = \min\{\ell, L - vt\}; \qquad s(t) = 0 \text{ for } t \geq t_r.$$

The no-wait arrival rates at the lane end are

$$A_E'(t) = ve(t), \quad A_A'(t) = va(t), \quad A_B'(t) = vb_0, \qquad t_f < t < t_r, \tag{9}$$

and zero outside that interval. These are arrivals to terminal service, not guaranteed complete departures when service binds. Ordinary total arrival rate is $v\rho_0$, with $\rho_0 = e_0 + a_0 + b_0$.

Before the front exits, remaining total mass is $s(t)\rho_0 = M$. During exit, it is $s(t)\rho_0$ and decreases by the total arrival rate. Including all terminal arrivals, conservation is

$$s(t)\rho_0 + A_E(t) + A_A(t) + A_B(t) = M. \tag{10}$$

For the ending lane in particular, early transferred mass is $\int_0^t s(u)m(u)\,du$, and

$$s(t)e(t) + A_E(t) + \int_0^t s(u)m(u)\,du = \ell e_0. \tag{11}$$

Hence the moving-footprint formulation does not lose mass when its occupied length shrinks. It also does not create an arbitrary current-stock discharge tail. Every surviving upstream element reaches the boundary according to its initial physical position and common travel clock.

6. A joint nonblocking terminal limit and its scientific conclusion

Let $R_A^{\text{out}}(v,t)$ and $R_B^{\text{out}}(v,t)$ be the declared downstream receiving-work rates. For a nonblocking terminal it is sufficient that

$$va(t) + \alpha ve(t) \le R_A^{\text{out}}(v,t), \qquad vb_0 \le R_B^{\text{out}}(v,t) \tag{12}$$

throughout the arrival window for all admitted actions. These are domain conditions, not externally supplied future receiving curves for prediction. As one existence member, keep declared constant finite downstream receiving rates $Q_A^{\text{out}}, Q_B^{\text{out}} > 0$ at an unconstrained exit. This retains the fixed-resource idea of S13 without adopting its simultaneous batch or proportional allocation. Conservative sufficient initial bounds are

$$v_+\alpha r \le Q_A^{\text{out}}, \qquad v_+ b_0 \le Q_B^{\text{out}}. \tag{13}$$

Indeed $a+\alpha e = r+(\alpha-1)e \le \alpha r$, and $v \le v_+$. Sufficiently sparse positive initial traffic can satisfy (A) and (13) together. This member has no terminal queue and no feedback that contradicts the upstream homogeneous clock. Allocation priority is immaterial when every demand is served; no new binding-terminal priority is selected by this limit.

Every initial position $x \in [0,\ell]$ completes at $(L-x)/v$. Its mean complete time and derivative are therefore

$$\overline{T}(v) = \frac{L-\ell/2}{v}, \qquad \frac{d\overline{T}}{dv} = -\frac{L-\ell/2}{v^2} < 0. \tag{14}$$

The preferred speed is the largest admissible one, irrespective of hidden merge composition. The result is a limiting/negative information-dependent control finding. It does not transfer S13's positive information/action certificate to the new member.

If (12) fails, terminal waiting is consequential. A queue on the ending lane or adjacent lane can change incoming flow and earlier receiving. Neither (3) nor (9) then automatically remains valid. A detached point queue with unlimited storage and no upstream influence would be a new approximation or new geometry assumption, not a source-supported default. This limitation does not prove that all compact homogeneous holding models are impossible.

For any eventual admitted terminal, retain the complete-population objective

$$F(v) = \int_0^{T_{\text{clear}}} N_{\text{unexited}}(t;v)\,dt, \qquad \overline{T} = F/M. \tag{15}$$

It counts upstream travel, any preparation waiting, terminal holding and all traffic until final exit. Staggered arrivals do not justify importing the half-clearance mean of a simultaneous proportional batch.

7. Information sufficiency and present disposition

With fixed known footprint and initial symmetry, aggregate mass M plus initial ending mass E_0 gives

$$e_0 = E_0/\ell, \qquad a_0 = b_0 = (M - E_0)/(2\ell).$$

Equation (4) then determines $e(t)$, and $a(t) = e_0 + a_0 - e(t)$ reconstructs the adjacent receiver. Thus one initial ending statistic is sufficient for the proposed upstream member. A strong aggregate-plus-ending-statistic rival can implement the identical equations and histories. Independent initial continuing-lane asymmetry would change that information contract; it is not admitted here.

Disposition. Homogeneous upstream closure is mathematically possible under explicit adaptations. A complete no-blocking member exists but loses the desired information-dependent speed contrast. A waiting-terminal member, applied MP6 domain, fresh reachable observer witness, derivative protocol and independent usefulness evidence remain open. The review stops at the original B2 system-interface gate; no simulator or experiment is authorized.

Source and adaptation record

- Carey, Balijepalli and Watling (2013), *Extending the Cell Transmission Model to Multiple Lanes and Lane-Changing*: diagram and local allocation ingredients, not this joined homogeneous task. Author manuscript, Sections 2, 4 and 5.
- Jin (2015), *Continuous formulations and analytical properties of the link transmission model*: initial/boundary-profile requirements, not one-stock closure with interior lateral sources. Primary preprint, Section 2.2.
- Jin (2021), *A link queue model of network traffic flow*: compact model-form rival with distinct internal dynamics. Primary preprint, introduction and conclusion.

All additional profile, rate-scale, uniform-source, speed-response and terminal assumptions are declared above. Conditional proof is not empirical calibration. The accepted S13 contract and its original scalar certificate are unchanged.